

Beyond Persona: Dynamic State Routing for Personal Chat-Log Replay

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Abstract

Personal digital twins built from conversational histories are often described in terms of persona, memory, or stylistic imitation, but these descriptions do not directly test whether a system can predict a user’s future conversational behavior. We introduce a proxy replay benchmark for personal GPT chat logs: given held-out contexts from a user’s historical conversation, predict the user’s next conversational action under leakage-free evaluation. After tightening a rule-based topic-shift label, a lexical nearest-neighbor baseline using CJK-aware character n-gram similarity achieves the highest point estimates on a fixed chronological split (0.35 accuracy, 0.38 macro-F1), while the learned state router scores 0.23 accuracy and 0.30 macro-F1. We do not detect a significant paired accuracy difference (McNemar $p = 0.074$), but bootstrap 95% CIs are wide (lexical macro-F1 [0.15, 0.53]) given $n = 40$ examples. A direct ablation shows that enabling state routing on top of the lexical baseline contributes zero additional signal — 40/40 predictions identical; this null result replicates across two pretrained embedding models (BGE-small-zh, MiniLM). Neither embedding model exceeds the CJK lexical tokenizer. The state-prior baseline confirms that dominant-action priors inflate accuracy while collapsing macro-F1. These results show that explicit state routing can appear useful under weak baselines but contributes nothing once lexical similarity is properly measured. The benchmark is single-user, rule-labeled, and not evidence of a complete human digital twin.

1 Introduction

Personal AI systems increasingly promise long-term memory, user adaptation, personalized dialogue, and digital-twin-like behavior [4, 2, 8, 11, 3, 9]. For such systems, it is not enough to produce a plausible description of a user or to imitate surface style. A useful personal agent should preserve behavioral continuity: when placed back into a historical context, it should predict what the user is likely to do next.

This paper studies that problem in the concrete setting of personal GPT chat logs. A long history of user–assistant interactions contains stable preferences, changing goals, unresolved tasks, and topic shifts. However, evaluating whether a model has captured those factors is difficult. Free-form response generation metrics are brittle; persona summaries are hard to verify; and using the full history in-context makes it unclear whether improvement comes from structured user modeling or raw retrieval. Similar concerns appear in personalization, social-simulation, and population-simulation benchmarks [5, 12, 7, 1]. We therefore focus on a narrower but measurable question: can a system predict the user’s next conversational action in held-out historical replay?

We propose a proxy replay benchmark for this setting. Each example consists of a conversation context, a target user turn, and a rule-derived action label such as *request how-to*, *compare options*, *provide more context*, or *topic shift*. The representation is explicit: persona facts, dynamic state, and memory nodes are materialized as intermediate objects rather than hidden inside a prompt. This setup lets us test which part of the representation is actually useful under held-out replay, and whether a lexical retrieval baseline is already stronger than a learned state router.

We organize the study around three questions. **RQ1**: does state-aware information improve held-out replay over a no-state baseline? **RQ2**: does a learned state selector improve over a heuristic state mapper? **RQ3**: does state routing add signal beyond strong lexical similarity? These questions evaluate a narrow proxy for behavioral continuity rather than broad human simulation.

Our main finding is nuanced. After tightening the proxy `topic_shift` label, a lexical context NN baseline using CJK-aware character n-gram similarity achieves the highest accuracy and macro-F1 (0.35 / 0.38),

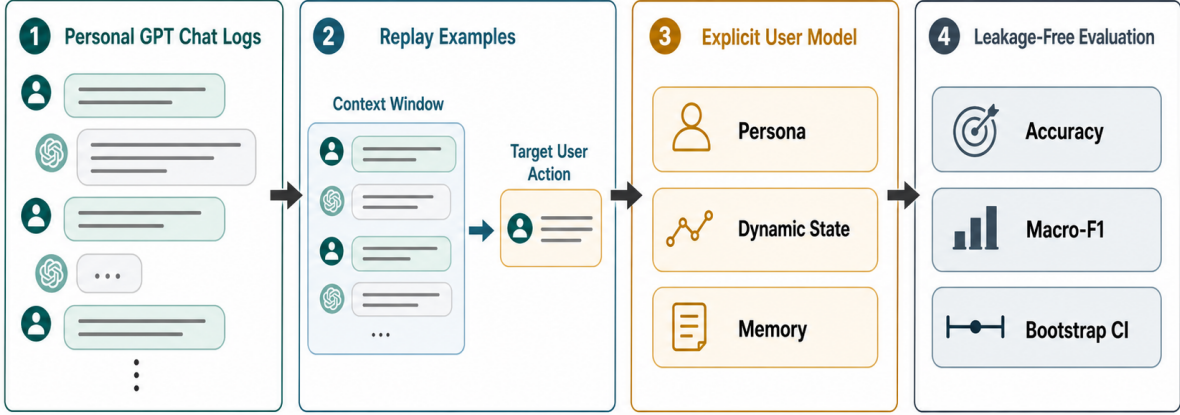


Figure 1: Paper overview: chat logs are normalized into replay examples, mapped into explicit user representations, and evaluated with leakage-free held-out metrics.

while the learned state router scores 0.23 / 0.30. We do not detect a significant paired accuracy difference (McNemar $p = 0.074$), but bootstrap CIs are wide at $n = 40$. Critically, adding state routing to the lexical baseline changes zero predictions (40/40 identical). State-only gains in earlier prototypes were artifacts of a weak similarity metric.

This paper makes four contributions:

1. We define a proxy next-action replay task for personal GPT chat logs, making behavioral continuity measurable.
2. We implement an explicit persona/state/memory pipeline and use leakage-free ablations to test the dynamic-state component.
3. We show that a CJK-aware lexical retrieval baseline performs best, exceeding both a learned state router and pretrained embedding models; state routing adds zero signal on top of lexical or embedding similarity.
4. We identify two pitfalls: state-prior baselines can inflate accuracy while collapsing macro-F1, and state-routing gains must be validated against strong lexical comparators with uncertainty estimates.

Figure 1 summarizes the replay setup and evaluation path.

The claim is deliberately scoped: the benchmark uses one user’s GPT logs with rule-derived proxy labels and a small held-out set ($n = 40$), so the results do not establish a complete digital twin or general human simulator. They show that state-aware replay is auditable, and that bootstrap CIs and paired tests are necessary for small replay benchmarks.

2 Related Work

Generative agents and behavior simulation. Generative agent systems use language models, memory, reflection, and planning to simulate believable behavior in interactive environments [4], and broader surveys organize this design space around perception, memory, planning, and action [6]. Our setting differs: rather than populating a sandbox with synthetic agents, we replay one real user’s historical GPT interactions and evaluate next-action prediction on held-out contexts.

Personalized language models. Personalization benchmarks such as LaMP evaluate whether language models can adapt to user-specific histories [5], and recent surveys distinguish personalization by granularity, technique, and evaluation protocol [10]. Most personalization work asks whether a model produces a better answer for a personalized task; our replay task asks whether an explicit user representation predicts the behavioral sequence of one user’s conversation.

Memory-augmented assistants and personal digital twins. Long-term memory systems for LLMs persist facts, preferences, and episodes across sessions. MemoryBank studies memory for companion-like interaction [11], MemGPT frames memory management as a virtual-context problem [3], and surveys emphasize that memory evaluation remains an open problem [9]. Digital-twin discussions share the ambition of persistent personal models but often mix memory, goals, and simulation into a broad identity claim. We align with this motivation but treat “digital twin” as a measurable replay problem, explicitly separating persona, dynamic state, and memory to test which component improves held-out behavior prediction.

User simulation and social interaction. User-simulation work produces realistic user behavior in dialogue, task, or social settings. SOTOPIA evaluates language agents in open-ended social interactions [12], with follow-up work training agents through interactive social learning [7]. Language models have also been used to simulate human samples in social-science settings [1]. Such simulators are valuable for evaluation and agent training, but they do not show that a model can replay a specific person’s historical trajectory. Our results support a related concern: static persona facts alone are not the dominant source of held-out replay gain, and a stronger semantic context comparator can exceed a learned state router.

Positioning. The closest existing directions provide believable agents, personalized task models, persistent memory, or simulated populations. This paper targets a different evaluation unit: one user’s historical conversation trajectory with proxy next-action labels. The contribution is not a general-purpose agent architecture, memory store, or population simulator. It is a state-aware replay formulation for personal chat logs with a controlled comparison showing that apparent state-routing gains can disappear under stronger lexical comparators. This deliberately conservative positioning is useful when a researcher wants an auditable proxy for personal behavioral continuity before making stronger claims about digital twins.

3 Replay Benchmark and Method

3.1 Replay task

Let a user’s chat history be segmented into episodes. Each replay example is a tuple (c_i, y_i, s_i) , where c_i is the context of recent user turns, y_i is the proxy next-action label of the target user turn, and s_i is an observed dynamic-state hint derived from the context. The prediction task is to infer y_i from c_i using only training examples and permitted state/profile information.

The action label space contains conversational actions such as asking for an example, asking why, comparing options, requesting implementation guidance, challenging or refining a prior answer, providing more context, asking a follow-up clarification, and shifting topic. Coarse labels group these actions into broader categories such as clarification, expansion, planning, comparison, and shift.

3.2 Pipeline

The pipeline has five stages. First, raw GPT export data is normalized into conversations, messages, sessions, episodes, and turns. Second, replay examples are built by selecting a context window and a target user turn from each episode. Third, proxy action labels and observed state hints are assigned with deterministic rules. Fourth, candidate predictors are trained or configured using only training examples. Fifth, each predictor is evaluated on held-out replay examples and divergences are recorded for error analysis. This structure is deliberately modular: label construction, state routing, persona extraction, and refinement can be audited separately.

3.3 Proxy labels

The benchmark uses real historical user messages, but the action labels are proxy labels. They are assigned by deterministic rules over the target user text and its relation to the previous user text. For example, text containing comparison markers is mapped to *compare options*; implementation markers map to *request how-to*; low similarity to the previous turn can indicate *topic shift*. This makes the benchmark reproducible, but

it also introduces label noise. We therefore treat the task as proxy replay evaluation, not as human-annotated intent recognition.

3.4 Explicit representation

The system materializes three kinds of user representation:

- **Persona facts:** relatively stable preferences or interaction habits inferred from training data.
- **Dynamic states:** context-specific states such as active problem, current goal, blocking issue, or current stance.
- **Memory nodes:** episodic or semantic records derived from historical conversations.

This explicit structure is useful because it exposes which signals are used. It also allows leakage checks: updates based on held-out examples must not be used to improve held-out predictions. The same separation also matches the broader memory-augmented-agent literature, where persistent memory is treated as a controllable state rather than a hidden side effect [3, 9].

3.5 Compared systems

Off. The no-state baseline predicts from nearest-neighbor similarity among training replay contexts using Jaccard word-overlap similarity without state hints.

Heuristic state hint. Maps marker-detected states to action-label hints, testing whether hand-written state cues help replay.

Learned state router. A Naive Bayes text classifier trained only on training contexts to predict a state type and value. The predicted state is mapped to an action hint and combined with the nearest-neighbor candidate list. This isolates the value of learning the state selector from context.

State prior. Predicts the training-set majority action for each observed state type/value. It is intentionally simple and strong on dominant modes, testing whether high accuracy can be obtained by state-action priors without balanced action modeling.

3.6 Training and inference protocol

All predictors are fit after the data split is fixed. For the learned router, the input is replay context text and the target is the rule-observed state type/value from training examples. At test time, the router predicts a state, converts it into an action hint, and re-ranks the nearest-neighbor candidates. The heuristic variant uses the same action-hint interface but obtains state hints from hand-written markers. The state-prior baseline removes context similarity entirely, predicting the most frequent training action for each observed state. This shared interface isolates how state is obtained and used.

The evaluation enforces two leakage barriers: held-out target turns are never used to construct training examples, and any profile, state-action mapping, or prior used at inference is estimated from training data only. Earlier refinement experiments leaked held-out information by updating the profile store while iterating over the full replay set; all paper claims use leakage-free settings and refinement is not claimed as a generalization mechanism.

4 Experiments

4.1 Data and metrics

The benchmark contains 257 replay examples from one user’s GPT chat history. The fixed chronological split has 179 train, 38 validation, and 40 test examples. After tightening the `topic_shift` relabel rule, the

Variant	Test Accuracy	Test Macro-F1
Off (no state)	0.0000	0.0000
Heuristic state hint	0.0750	0.0858
State prior (majority per state)	0.2750	0.0816
Learned state router	0.2250	0.3039
Logistic regression (TF-IDF)	0.3000	0.1729
Lexical context NN (CJK n-gram)	0.3500	0.3839

Table 1: Fixed held-out replay results ($n = 40$). All methods are leakage-free. 95% CIs: lexical acc [0.20, 0.50], F1 [0.15, 0.53]; learned acc [0.10, 0.35], F1 [0.09, 0.43]; LR acc [0.18, 0.45], F1 [0.07, 0.28].

Variant	Test Accuracy	Test Macro-F1
Lexical context NN (no state)	0.3500	0.3839
Lexical + learned state router	0.3500	0.3839

Table 2: State routing ablation. Enabling the learned router produces identical predictions (40/40), so state hints add no marginal signal beyond CJK-aware lexical similarity.

fixed test split remains imbalanced: 17 *follow-up clarification*, 12 *topic shift*, 3 *request how-to*, 3 *compare options*, 2 *provide more context*, 2 *ask for example*, and 1 *ask why*. We also evaluate five random episode splits and five random conversation splits.

We report accuracy and macro-F1. Accuracy measures overall action prediction, but macro-F1 is the primary decision metric because the label distribution is imbalanced and minority actions matter for replay fidelity. Given the small test set ($n = 40$), we compute bootstrap 95% confidence intervals (10 000 resamples) and McNemar’s test with Yates’ correction for key pairwise comparisons.

4.2 Main fixed-split results

Table 1 shows fixed held-out results with bootstrap 95% CIs. The Off baseline scores zero because chronological train/test periods differ and Jaccard word overlap lacks CJK-aware tokenization or state hints. The lexical context NN achieves the highest accuracy and macro-F1. We do not detect a significant paired accuracy difference from the learned router (McNemar $\chi^2 = 3.20$, $p = 0.074$), but this is not a non-inferiority result and CIs are wide.

4.3 State routing ablation

We test whether state routing adds signal beyond lexical similarity by running the lexical baseline with and without the learned router. Table 2 shows the central result: predictions are 40/40 identical. State hints are either already implied by CJK-aware context similarity or too weak to override nearest-neighbor rankings; earlier state-only gains were artifacts of weak Jaccard word overlap.

4.4 Embedding baselines

We test whether sentence-transformer embeddings outperform the CJK-aware lexical tokenizer. BGE-small-zh-v1.5 and MiniLM-L12-v2 use cosine similarity on L2-normalized embeddings with the same k-NN voting and state-routing protocol as the lexical baseline.

Neither embedding model exceeds the CJK lexical baseline: MiniLM matches accuracy (0.35) but has much lower macro-F1 (0.16 vs. 0.38), while BGE underperforms on both metrics. State routing again adds zero signal (40/40 identical for both models), so the bottleneck is not simply the absence of pretrained semantics; a well-tuned domain tokenizer already captures much of the relevant similarity structure.

4.5 Random split sensitivity

Table 4 reports mean and standard deviation across five random episode and five random conversation splits. Lexical context retrieval has the best macro-F1 point estimates, while gaps are small under conversation

Variant	Dim	Accuracy	Macro-F1
Lexical CJK n-gram (production)	—	0.3500	0.3839
BGE-small-zh-v1.5	512	0.3000	0.0659
+ state router	512	0.3000	0.0659
MiniLM-L12-v2	384	0.3500	0.1602
+ state router	384	0.3500	0.1602

Table 3: Embedding baselines vs. lexical baseline. Both embedding models match or underperform the CJK lexical tokenizer, and state routing adds zero marginal signal (40/40 identical for both models).

Split strategy	Variant	Accuracy	Macro-F1
Episode	Off	0.1390 $\pm$ 0.1019	0.1065 $\pm$ 0.0615
Episode	Heuristic	0.2214 $\pm$ 0.1559	0.2142 $\pm$ 0.0765
Episode	Learned	0.3669 $\pm$ 0.1202	0.2523 $\pm$ 0.0485
Episode	Lexical NN	0.3754 $\pm$ 0.1029	0.3102 $\pm$ 0.0597
Episode	State prior	0.2885 $\pm$ 0.0964	0.0790 $\pm$ 0.0356
Conversation	Off	0.2139 $\pm$ 0.0971	0.1628 $\pm$ 0.0588
Conversation	Heuristic	0.3144 $\pm$ 0.1119	0.2383 $\pm$ 0.0508
Conversation	Learned	0.4035 $\pm$ 0.1644	0.3035 $\pm$ 0.0895
Conversation	Lexical NN	0.4043 $\pm$ 0.1443	0.3164 $\pm$ 0.0742
Conversation	State prior	0.2040 $\pm$ 0.0932	0.0593 $\pm$ 0.0364

Table 4: Random split sensitivity. Lexical context retrieval has the highest macro-F1 point estimates after tightening the proxy labels, but variance is high and only five splits are used.

splits. The state-prior baseline obtains moderate accuracy but very low macro-F1, indicating collapse toward dominant labels.

4.6 Interpretation and validity

Six conclusions follow. First, lexical retrieval has the highest point estimates across split strategies and similarity methods. Second, we do not detect a significant paired accuracy gap between learned routing and lexical retrieval at $n = 40$ (McNemar $\chi^2 = 3.20$, $p = 0.074$), but uncertainty is large. Third, state routing adds no marginal signal on top of lexical retrieval. Fourth, this null result replicates for both embedding models. Fifth, logistic regression falls between state-only methods and lexical NN. Sixth, wide bootstrap CIs show that point estimates are fragile at this scale.

Episode splits reduce dependence on one favorable chronology, while conversation splits move whole conversations between train and test. Lexical retrieval remains strong under both. The state-prior sanity check mainly improves accuracy (0.275) while macro-F1 stays near zero (0.082), consistent with majority-mode memorization. Reproducibility artifacts are included in the evaluation module.

5 Analysis

5.1 Where does learned routing help?

Table 5 summarizes the fixed-split diagnosis after tightening the relabel rule. Rows with extremely small counts are omitted. The main change is that the old `topic_shift` bucket no longer swallows same-thread follow-ups. Once those boundary cases are separated out, the learned router matches or exceeds the lexical NN on `compare_options` and `follow_up_clarification`. The lexical baseline, however, catches more signal on `topic_shift`, `request_how_to`, and `ask_for_example`, which gives it the overall edge.

5.2 Why does state routing add nothing to stronger similarity?

The ablation in Table 2 shows zero marginal gain from state routing on lexical retrieval, and Table 3 replicates this for both embedding models. Two mechanisms explain this. First, CJK-aware character n-gram similarity

	Off	Heuristic	Learned	Lexical NN
<i>Per action-label accuracy</i>				
topic_shift	0.000	0.083	0.417	0.667
request_how_to	0.000	0.000	0.000	0.667
compare_options	0.000	0.000	0.333	0.333
follow_up_clarification	0.000	0.000	0.118	0.059
ask_for_example	0.000	0.000	0.000	0.500
<i>Per state-type accuracy</i>				
active_problem	0.000	0.045	0.273	0.273
blocking_issue	0.000	0.333	0.333	0.333
current_stance	0.000	0.050	0.200	0.250

Table 5: Fixed-split diagnosis after relabel tightening, broken down by action label and by state type. Counts are small for some rows; this table is diagnostic rather than definitive. Corrected numbers from comprehensive re-evaluation.

already captures topic continuity and shift patterns that the state router was designed to detect; pretrained embeddings do not add orthogonal signal at $n = 40$. Second, the router is trained on the same data as the k-NN, making its predictions redundant once the similarity metric is strong. To contribute independent value, state routing would need to capture long-range goals or user preferences absent from surface similarity, but those signals are either absent or too noisy at this scale.

Notably, the learned state router alone has a nontrivial macro-F1 point estimate (0.304 vs. 0.384 for lexical retrieval), so the problem is not that state routing has no signal under this proxy task — it is that state routing adds nothing *beyond* what CJK-aware lexical similarity already captures.

5.3 What remains unsolved

The method remains weak on minority action labels such as *follow-up clarification* and *provide more context*, where user intent is mixed with continuation or background elaboration. These labels are the most brittle part of the proxy space. The state-prior baseline further shows why balanced metrics matter: it obtains moderate accuracy (0.275) by selecting the majority action for each state, but its macro-F1 (0.082) is far lower than either the learned router or the lexical baseline because it fails on minority labels entirely.

The project also includes a discrepancy-driven refinement mechanism, but leakage-free ablation shows that train-only refinement does not improve held-out performance. This prevents an overclaim that iterative profile updates solve generalization. Refinement is best treated as a debugging and error-analysis tool, not as the main paper method.

6 Limitations and Ethics

Proxy labels. The benchmark uses rule-derived action labels rather than human annotations, making evaluation reproducible but introducing label noise. A manual spot audit found the dominant noise mode is `topic_shift` overuse on same-thread refinements. Until a full manual label audit is available, results should be interpreted as evidence about a reproducible proxy task rather than a validated model of user intent.

Single-user scope. Main results are from one user’s GPT chat history, so they do not establish cross-user generality. As a preliminary stress test, we converted PERSONA-Bench Reddit threads into 135k proxy replay examples over 2,871 authors. Majority prediction again inflated accuracy (0.41 accuracy, 0.06 macro-F1 on a 5k test cap), while TF-IDF LR was more balanced (0.23/0.14) and a larger GPU MiniLM nearest-neighbor run on 50k/10k examples reached 0.30/0.13. A 90-example stratified audit found 78 visibly supported and 12 borderline proxy labels. These results support the metric caveat, but not full method generalization.

Imbalanced labels. The test split is dominated by topic shifts and follow-up clarifications. We emphasize macro-F1, but minority-label performance remains weak. Future work should improve label design or evaluate behavior chains.

Privacy and scope. Personal chat logs can contain sensitive information. Any deployment must use explicit consent, local processing, redaction, and access control. The term “digital twin” is only a motivating direction, not a claim of complete human simulation.

7 Conclusion

We presented a proxy replay benchmark for personal GPT chat logs and evaluated explicit dynamic-state routing for next-action prediction within a broader persona/state/memory pipeline. After tightening the `topic_shift` relabel rule, a CJK-aware lexical context baseline achieves the highest point estimates (0.35 accuracy, 0.38 macro-F1), while the learned state router scores 0.23 accuracy and 0.30 macro-F1; we do not detect a significant paired accuracy difference at $n = 40$ (McNemar $p = 0.074$). A direct ablation shows that state routing contributes zero additional signal on top of lexical retrieval (40/40 identical predictions), and this null result replicates across BGE-small-zh and MiniLM. The state-prior baseline confirms that balanced metrics are necessary.

The broader implication is methodological: personal digital-twin research needs measurable replay tasks before making strong user-fidelity claims, and state-aware methods must be evaluated against strong lexical comparators with uncertainty estimates. Future work should add stronger multi-user runs, behavior-chain metrics, and better calibrated models.

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